Mini Project 3

Amazon Reviews

Text sentiment analysis

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01

Overview

A high-level summary & purpose.

Overview

Context

The sentiment analysis focus on a dataset on **Amazon customer reviews** in the **Toy & Games** category. Reviews are collected from 1996 to 2023. The original dataset contains over 16 million reviews where we take a sample of 20,000 reviews for the purposes of the analysis.

Outcome

Negative reviews = 0 | Positive Reviews = 1

Question

How well does review text predict customer satisfaction/rating?

Business Case

By predicting text sentiment can we identify misleading star ratings and utilise text components to have a far better picture?

- Prevent trust erosion
- Churn risk of customers
- Missed product issues or complaints

02

Data Background

A review of the data.

Data Background

20,000

Records

11

Features

Feature Name	Data Type	Description
rating	integer	Rating from respondent between 1 to 5
title	string	Title of the review short description
text	string	Long form response
images	list	url address of product image/s
asin	string	Amazon product code / identifier
parent_asin	string	Amazon parent product code / identifier
user_id	string	Id of respondent/customer
timestamp	Date time	Date and time review was made
helpful_vote	integer	Up votes of how helpful by other users
verified	boolean	A true false of whether the user is a purchaser of product
Output	**boolean**	**0 = negative, 1 = positive (4-5 ratings)**

Title + Text columns – text stats

output	Avg Word Count	Average Reading Time	Text Level
Negative	43.0	2m 39s	7th and 8th grade
Positive	33.0	2m 0s	8th and 9th grade

Additional notes

- Combined title and text column to a single column called total review. After cleaning and preperation a new feature is created called **clean_text** for modelling
- Output column is created by grouping ratings 1-3 as negative and ratings 4 & 5 as positive.

03

EDA & Text Prep

Exploring the data and text preparation.

EDA & Text Prep

Text cleaning & Preparation

Text Cleaning

- Dropped features:-
  - title
  - product code
  - parent product code
  - reviewer id
  - time stamp
  - helpful votes
  - verified purchase
  - year
- Manual text processing
  - Expand contractions
  - Removed html tags with Beautiful soup
  - Emoticons, symbols, pictographs, flags, dingbats via Unicode
  - Normalise case
  - Removed hyphens, spaces, digits, non-ASCII characters

Text Preparation

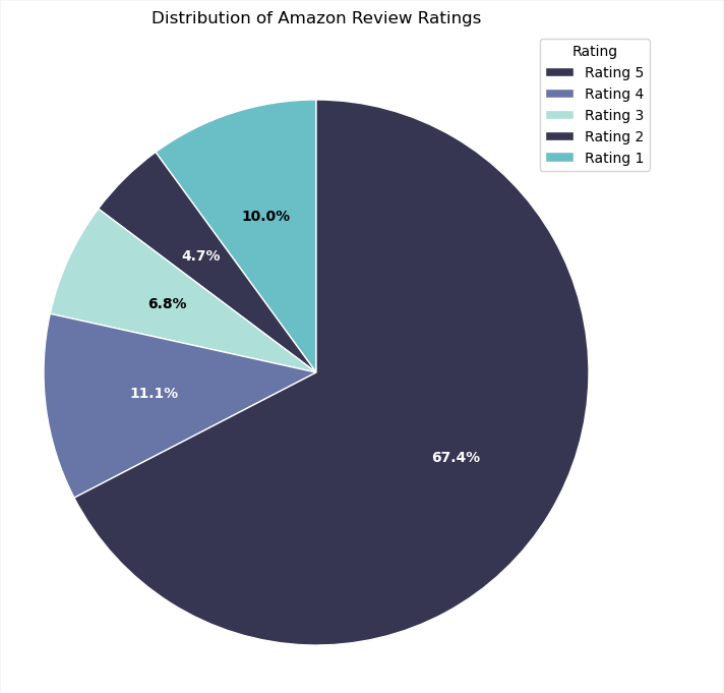
- Spacy
  - Stop words
  - Lemmatisation
  - Punctuations

Initial Findings

- There are a subset of customer/respondents who are regular reviewers.
- Parent asin products have multiple reviews though distribution is somewhat evenly spread.
- Majority of reviews had zero helpful votes.
- Review growth started to grow from **2010** to **2023**. Distribution of reviews lean towards more recent years.

EDA & Text Prep

Visualising



TextBlob - Positive polarity

rating	text
5.0	Perfect for the luau bridal shower I hosted for my friend.
5.0	They do stretch out once they have been snapped and worn a few times, but they are perfect for a kids party give away.
4.0	I teach Kindergarten and this is perfect for my word wall. The pictures with the letters help the children with their letters.
5.0	These are perfect to add embelishments for projects. My daughter and I have made several head bands. I would recommend this item.
4.0	excellent

TextBlob - Negative polarity

rating	text
1.0	Terrible experience The price dropped $10 before I received it
1.0	Does not spray and kids couldn't slide.
1.0	Horrible - glue gun used to make these?
1.0	It doesn't even work
3.0	The balloons smelled horrible!

Other notes

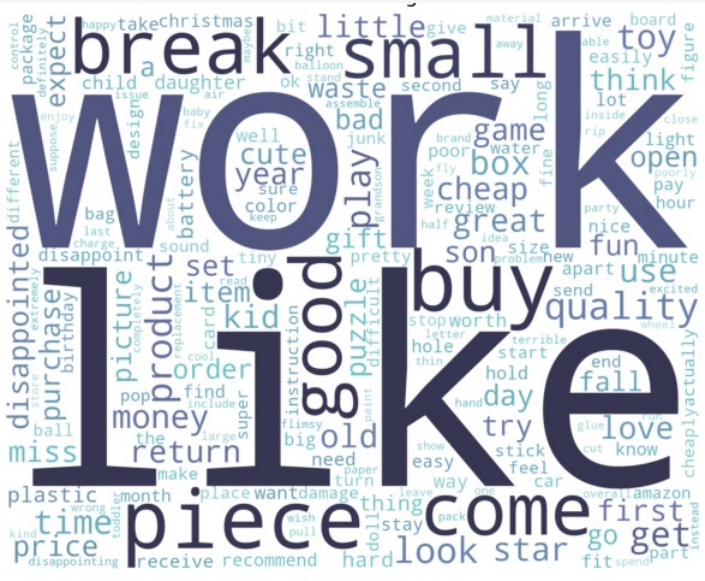
Outcome column created to show

- 78.5% - Positive
- 21.5% - Negative

EDA & Text Prep

Word Cloud

Negative - 0



Positive - 1

04

Modelling

A comparison of different models and their performance.

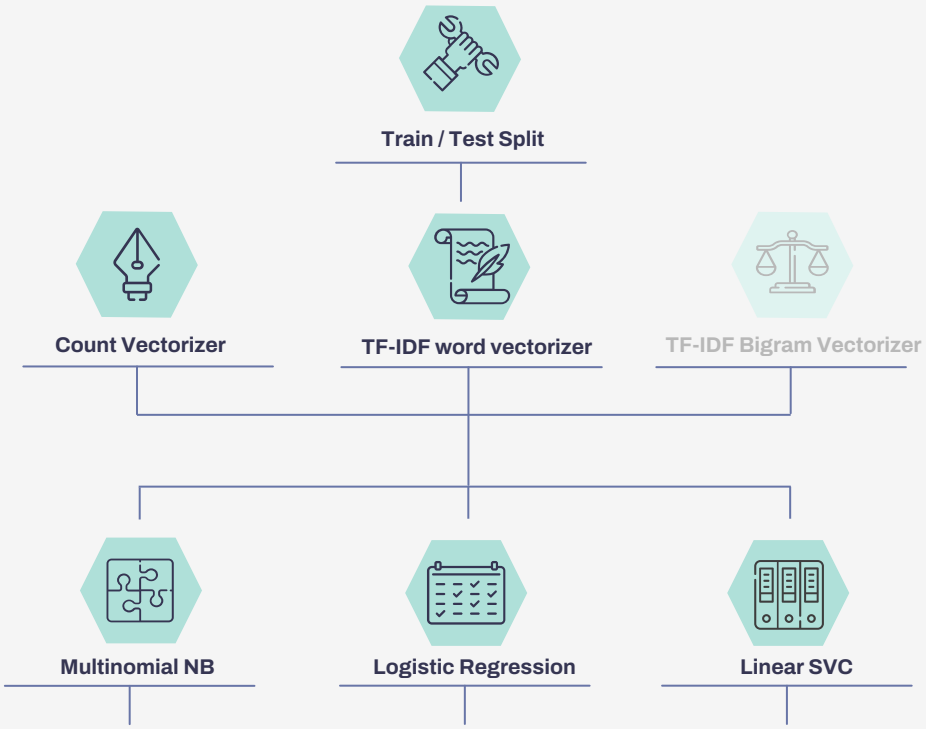
Modelling

Building the modelling

Process

- Vectorised the text using **Count** and **TF-IDF** method. Unigrams and bigrams
- Selected three models to run default baseline.
- Review any additional text cleaning.
- Select best model.

Vectorizer	Features	Non-zero entries	Density
Count (Unigram)	13987	212769	0.000952
TF-IDF (Unigram)	13987	212769	0.000952
TF-IDF (Bigram)	351409	451266	0.000080



Modelling

Model Comparison

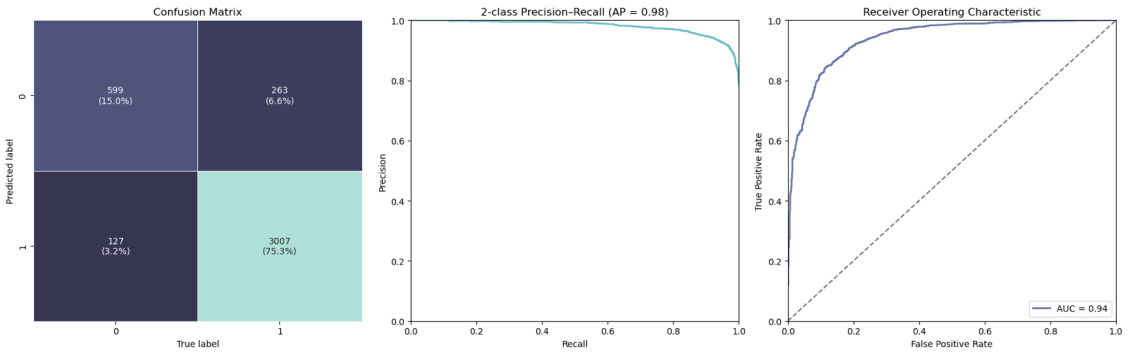
Multinomial Performance

Vectorizer + Model	Accuracy	Precision	Recall	F1
TF-IDF (uni) + LinearSVC	0.9024	0.8992	0.9024	0.8993
Count (uni) + LogReg	0.9024	0.8992	0.9024	0.8991
TF-IDF (uni) + LogReg	0.9014	0.8999	0.9014	0.8950
Count (uni) + MultinomialNB	0.8961	0.8924	0.8961	0.8922
Count (uni) + LinearSVC	0.8866	0.8836	0.8866	0.8847
TF-IDF (uni) + MultinomialNB	0.8391	0.8640	0.8391	0.7994

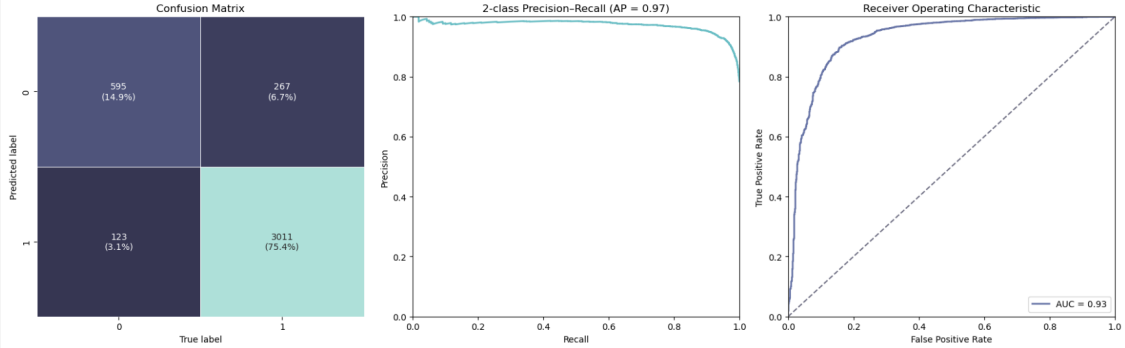
Words driving Predictions (Linear SVC)

Positive feature	Positive coefficient	Negative feature	Negative coefficient
love	3.5532	disappointed	-3.9993
perfect	3.4199	disappointing	-3.0341
awesome	2.6301	poor	-2.6885
great	2.5817	return	-2.5666
perfectly	2.0071	waste	-2.3796
fantastic	1.9300	cheaply	-2.2184
adorable	1.6989	meh	-2.2134
nicely	1.6683	break	-1.9598
exactly	1.6385	poorly	-1.9121
highly	1.5926	junk	-1.9046
amazing	1.5334	frustrating	-1.8841
beautiful	1.5101	horrible	-1.8643
value	1.5016	mislead	-1.7463
solid	1.4920	sadly	-1.7396
fun	1.4484	barely	-1.6978

TF-IDF (uni) + Linear SVC



Count (uni) + LogReg



05

Conclusion

A recap on findings and key points

Conclusion

Conclusion

- Identified **TF-IDF unigram** combined with a **LinearSVC** model to be the best performing. Scoring best on all metrics including our primary **F1 Score**.
- It has also been able to resolve some of our original business case questions. Misleading reviews that may reveal more.

Question

- How well does review text predict customer satisfaction/rating?

Future work

- More work with bigrams and feature reduction to improve performance.
- Further feature engineering.
- Further tuning of the model maybe needed or testing recall over precision.

total_review	rating
needs air the pamphlet included talks about is thr sand moves to quick or to slow and a regulator on how to fix it but unfortunately this is not included and mine tends to move to fast the smaller versions are more disappointing than larger versions which is sad as it is mesmorong and calming and great for desk environment if it works	4.0
my son loves this and it is not cheaply made my son loves this and it is not cheaply made we bought him a different brand before and it did not last a day as he is really rough on his toys this is still going strong months later	5.0
bath time is more fun now the toys are just as pictured bright rubber animals my baby months old likes them but she was frustrated at first as she had hard time holding them now she learned to use both her little hands and does not complain anymore the only thing i may not like is a little chemical smell but i hope it is not harmful and will disappear soon	4.0
looks great looks just like the picture we like this version better than the with the fire and ugly dresslol	5.0
not messy i am a big fan of play doh so i decided to give their slime pack a try my year old daughter said this slime is the least sticky and messy	5.0

Thank You

Questions?